

Chapter

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Introduction and Pascal's law

Fluid static:

Fluid statics is the study of fluid behavior at rest. It has many practical applications in various fields, including civil engineering, naval architecture, petroleum engineering, aerospace engineering, and biomedical engineering.

Fluid statics is a fundamental branch of engineering that is essential for the design and operation of many different types of systems and devices.

Fluid:

A fluid is a substance that flows or deforms when subjected to shear stress. This means that a fluid cannot resist a force that tends to deform it. Liquids and gases are both fluids.

- **In liquid:**
Fluids under pressure can perform work, or they can carry nutrients and essential soluble, like blood flowing through our arteries and veins.

The study of fluids is subdivided into two parts

Hydrostatics or fluid Statics: The study of fluid at rest.

Fluid Dynamics: The study of properties and effect of moving fluid

- **In Gases:**
Flowing gases cause pressure difference that can lift a massive cargo plane or roof off a house in a hurricane.

Properties of Gases and Liquids:

Property	Gas	Liquid
Shape	No fixed shape	Takes the shape of its container
Volume	No fixed volume, expands to fill its container	Definite volume that does not change significantly with pressure
Compressibility	Highly compressible	Relatively incompressible
Density	Low density	Higher density than gases but lower density than solids
Diffusion	Exhibits rapid diffusion and mixes uniformly with other gases	Does not diffuse as quickly as gases but still mixes uniformly over time
Viscosity	Low viscosity, flows easily	Exhibits viscosity, flows, but its flow rate is slower than gases

Pascal experimentally observed that an external force when applied on the surface of a liquid increases the liquid pressure at the surface of the liquid. This increase in liquid pressure is transmitted equally in all directions and to the wall of the container. This result is called Pascal's law.

Statement:

Pascal States that:

"When a change in pressure is applied at any point to any static fluid, it is transmitted perfectly to all portions of the fluid and to the wall of the container"

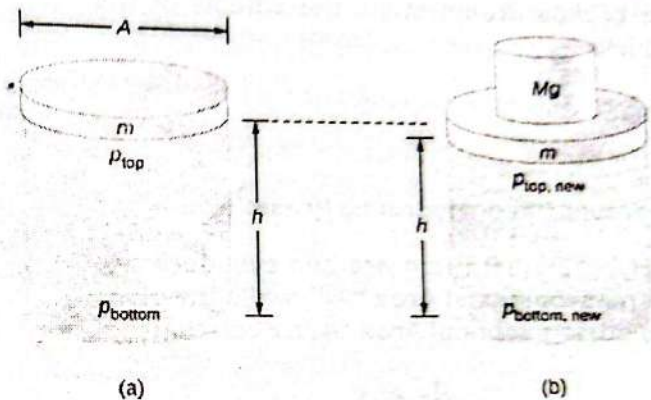
Explanation:

Suppose a cylindrical vessel or container of cross sectional Area "A" having height "h" of movable piston of mass "m" as shown in the diagram

Adding weight Mg at the top of the piston as additional weight also act over the top of the liquid, so

$$P = \frac{F}{A} = \frac{W}{A}$$

$$\Delta P_{top} = \frac{Mg}{A}$$



According to Pascal's law, the pressure at all points in the fluid changes by the same amount i.e. Mg. Thus the pressure at the bottom of the container is equal to the sum of atmospheric pressure, the pressure due to the fluid and the pressure supplied by the mass.

So, the change in pressure at the bottom of the container due to the mass is written as,

$$\Delta P_{bottom} = \frac{Mg}{A}$$

By comparing we get

$$\Delta P_{top} = \Delta P_{bottom}$$

Conclusion:

From the above we conclude that as the changes in Pressure are the same everywhere in the

Fluid, so we may write also

$$\Delta P = \Delta P_{top} = \Delta P_{bottom} = \Delta P_{everywhere}.$$

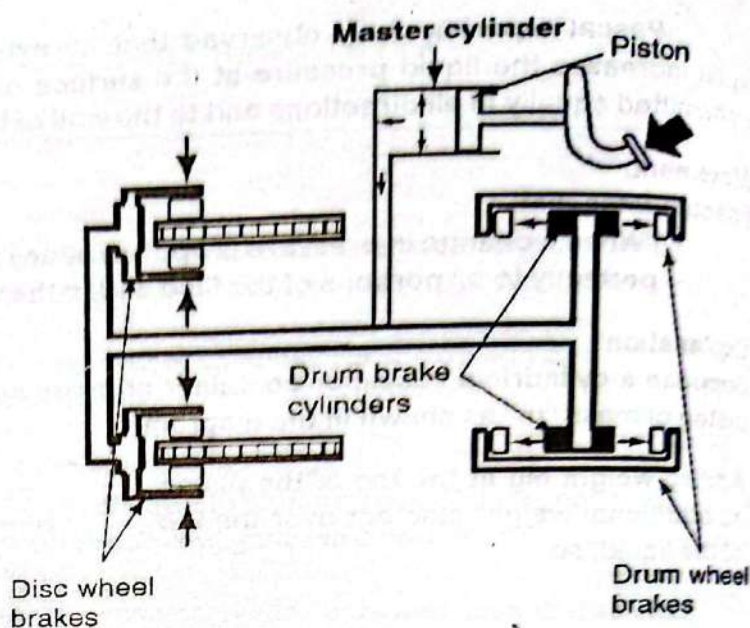
Applications:

Pascal's law finds numerous applications in our daily life such as automobiles, hydraulic brake system, hydraulic jack, hydraulic press and other hydraulic machines etc. Here we explain briefly the following

- Automobile Hydraulic Brake System
- Hydraulic lift or Hydraulic Jack

Automobile Hydraulic Brake System:

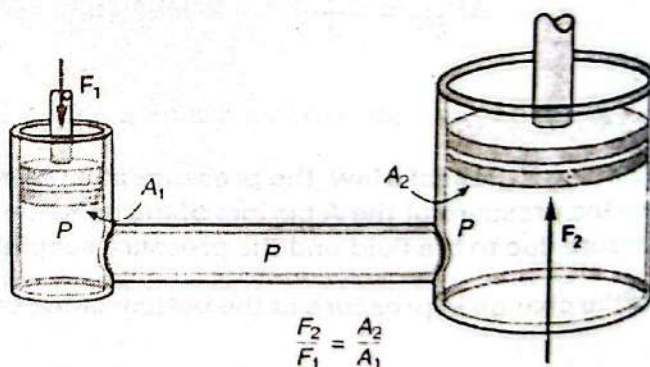
The braking system of cars, buses etc. works on the principle of Pascal's law. The hydraulic brakes allow equal pressure to be transmitted throughout the liquid. When brake pedal is pushed, it exerts force on the master cylinder which increases the liquid pressure in it. The liquid pressure is transmitted equally through the liquid in the metal pipes to all the pistons of other cylinders. Due to the increase in liquid pressure, the piston in the cylinder move outward pressing the brake pedal with the brake drums. The force of friction between the brakes drums stops the wheels of the vehicle.

**Hydraulic Lift or Hydraulic Press:**

In Hydraulic lift there are two cylinders one of cross-sectional Area " A_1 ", while the other has cross-sectional Area " A_2 ", Such that

$$A_2 > A_1$$

The cylinders, each of which is fitted with a piston are connected by a tube and filled with a suitable fluid usually oil called Hydraulic fluid. Initially the pistons are at the Same level and exposed to the atmosphere.



Now suppose we push down smaller piston with the force F_1 . The pressure P_1 exerted by this piston is:

$$P_1 = \frac{F_1}{A_1}$$

Similarly, the pressure on the piston lifting the vehicle is P_2 , which can be written as,

$$P_2 = \frac{F_2}{A_2}$$

By Pascal's law, the pressure on the large piston is same as the pressure on the smaller piston.

$$P_2 = P_1$$

Putting the values from above equations we get:

$$\begin{aligned}\frac{F_2}{A_2} &= \frac{F_1}{A_1} \\ F_2 &= \frac{A_2 \times F_1}{A_1} \\ F_2 &= \frac{A_2 F_1}{A_1}\end{aligned}$$

OR

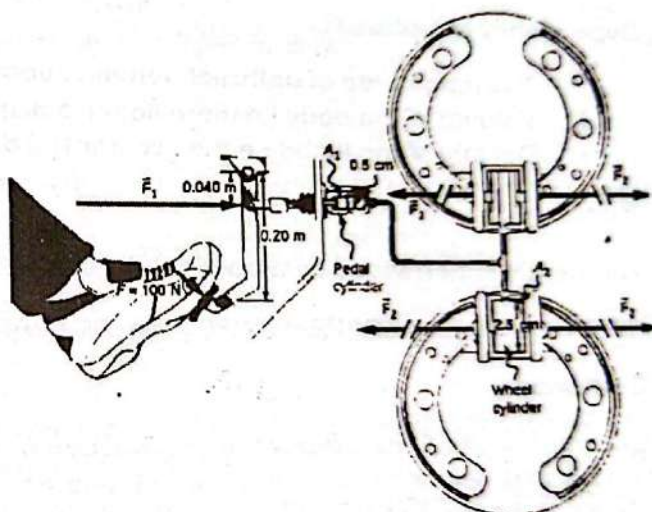
$$\frac{F_2}{F_1} = \frac{A_2}{A_1}$$

Where,

Mechanical Advantage (M.A) is a measure of the ratio of output force to input force in a system $= \frac{F_2}{F_1}$. Depending on the ratio $\frac{A_2}{A_1}$ which is the mechanical advantage, the force F_2 can be as large as required.

Example:

Consider the automobile hydraulic system as shown. Suppose a force of 100 N is applied to the brake pedal, which acts on the pedal cylinder (acting as a "master" cylinder) through a lever. A force of 500 N is exerted on the pedal cylinder. Pressure created in the pedal cylinder is transmitted to the four wheel cylinders. The pedal cylinder has a diameter of 0.500 cm and each wheel cylinder has a diameter of 2.50 cm. Calculate the magnitude of the force F_2 created at each of the wheel cylinders.



Explanation:

We are given the force F_1 applied to the pedal cylinder. The cross-sectional areas A_1 and A_2 can be calculated from their given diameters. Then we can use the following relationship to find the force F_2 .

$$\frac{F_2}{F_1} = \frac{A_2}{A_1}$$

Solution

Pascal's principle applied to hydraulic systems is given by :

$$\frac{F_2}{F_1} = \frac{A_2}{A_1}$$

$$F_2 = \frac{A_2}{A_1} F_1 = \frac{\pi r_2^2}{\pi r_1^2} F_1$$

$$F_2 = \frac{\pi (1.25)^2}{\pi (0.250)^2} \times 500 = 1.25 \times 10^4 \text{ N}$$

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Archimedes' Principle

ARCHIMEDES' PRINCIPLE:

Introduction:

We might have noticed that a mug filled with water feels light under water but feels heavy as soon as we take it out of water. More than two thousand years ago, the Greek scientist, Archimedes noticed that there is an upward force which acts on an object kept inside a liquid. This upward force is called as the upthrust of the liquid.

Statement:

"When an object is totally or partially immersed in a liquid, an upward buoyant force acts on it, which is equal to the weight of the liquid it displaces".

Dependency of Upthrust:

The magnitude of upthrust depends upon the followings.

- Volume of the body i.e more liquid is displaced, the greater the upthrust.
- Density of the liquid i.e the greater the density, the greater is the upthrust

Therefore,

The weight experienced by a body totally submerged is always less than its actual weight,

$$\text{APPARENT WEIGHT} = \text{ACTUAL WEIGHT} - F_{\text{upthrust}} \text{ (or UPTHURST)}$$

Derivation:

Consider a solid cylinder of cross-sectional area A and height " h " immersed in a liquid as shown in figure. Let h_1 and h_2 are the depths on the top and bottom faces of the cylinder respectively from the surface. Then,

$$h_2 - h_1 = h$$

If P_1 is the liquid pressure at depth h_1 , and ρ is the density of the liquid then

$$P_1 = \frac{F_1}{A} = \frac{mg}{A}$$

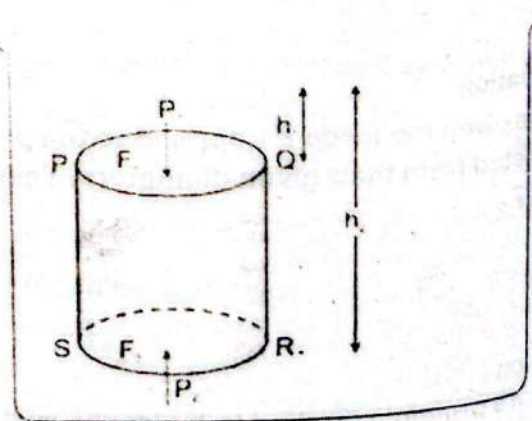
Where m is the mass of the liquid of the liquid

Mass of liquid = volume $\times$ density

$$m = (A \times h_1) \times \rho$$

$$\therefore \rho = \frac{m}{V}$$

Putting the value of m in equation P_1



$$P_1 = \frac{Ah_1\rho g}{A}$$

$$\therefore P_1 = \rho gh_1$$

$$P_2 = \rho gh_2$$

Similarly,

Let F_1 be the force exerted at the cylinder top by the liquid due to pressure P_1 and the force F_2 is the force exerted at the bottom of the cylinder by the liquid due to P_2 then,

$$F_1 = P_1 A = \rho gh_1 A$$

$$F_2 = P_2 A = \rho gh_2 A$$

F_1 and F_2 are acting on the opposite faces of the cylinder. Therefore, the net force $F_{upthrust}$ will be $F_2 - F_1$ in the direction of F_2 as $h_2 > h_1$

This net force $F_{upthrust}$ is called the up thrust of the liquid.

$$\therefore F_2 - F_1 = \rho gh_2 A - \rho gh_1 A$$

$$F_{upthrust} = \rho g A (h_2 - h_1) \because h_2 - h_1 = h$$

or Upthrust of liquid is $= F = \rho g Ah$

$$\therefore Ah = V$$

Eq: becomes,

$$\text{Upthrust} = F_{upthrust} = \rho g V$$

Here Ah is the volume (V) of the cylinder and is equal to the volume of the liquid displaced by the cylinder.

Note: The Equation shows that the upthrust acts which on the body immersed in a liquid and is equal to the weight of liquid displaced, which is Archimedes principle.

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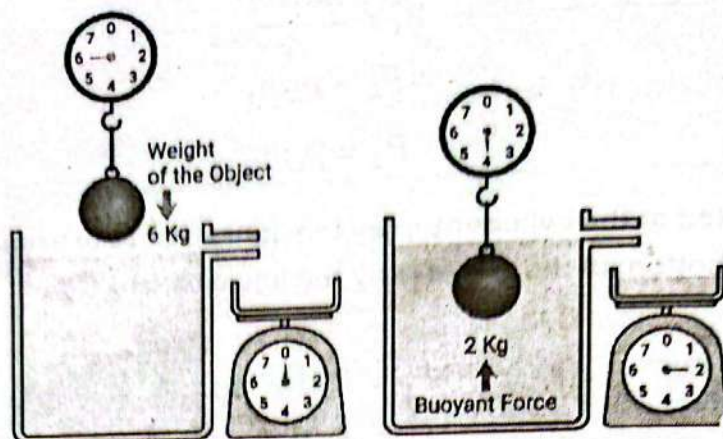
Buoyant Force

When an object is immersed in a fluid the pressure on the lower surface of the object is higher than the pressure on the upper surface. The difference in Pressure Leads to an upward force acting on the object due to the fluid Pressure called buoyant force F_B or Up thrust Force $F_{upthrust}$ and the phenomena is called buoyancy.

BUOYANT FORCE (F_B):

The buoyant force acting on an object fully or partially submerged in a fluid is equal to the weight of the fluid displaced by the object.

$$F_B \equiv W(\text{displaced fluid}).$$



Note that F_B is calculated by using Archimedes Principle

Buoyancy in Liquids:

- If the weight of the submerged object is greater than buoyant force ($W > F_B$), the object sinks.
- If the weight of the submerged object is less than the buoyant force ($W < F_B$), the object rises to the surface and floats.
- If the weight of the object is equal to the buoyant force ($W = F_B$), then the object remains at any level in the liquid; it neither sinks nor rises to the surface.

Buoyancy in Gases:

Archimedes' principle is also applicable for gases. For gases the principle is stated as:

"An object surrounded by air is buoyed up by a force equal to the weight of the air displaced"

LAW OF FLOATATION:

In case of floating object, the object may be partially immersed. The upthrust is always equal to the weight of the fluid displaced by the object. This is the principle or law of flotation. It states that

Statement:

"A floating object displaces a fluid having weight equal to the weight of the object."

Condition of object to Float:

- The average density of the object should be less than the density of the fluid in which the object has to be floating.

Example:

A ship is very heavy but it floats because it is hollow inside it contains air, this causes its average density be lower than that of water.

- The upthrust on the object must be equal to the total weight of the object

Example:

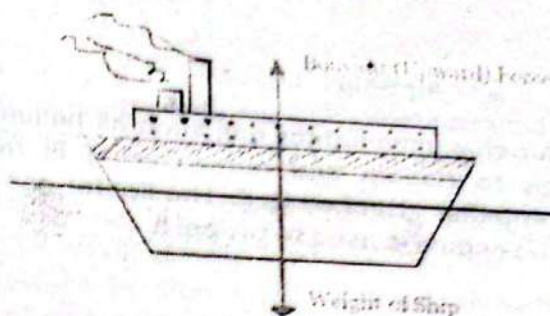
A coin will sink to the bottom when placed on the Surface of water, this is because the upthrust of water on coin is less than its weight (coin weight).

- The volume of object submerged must be large so as to displace large amount of water.

Example: Ship, wooden log.

• Ships:

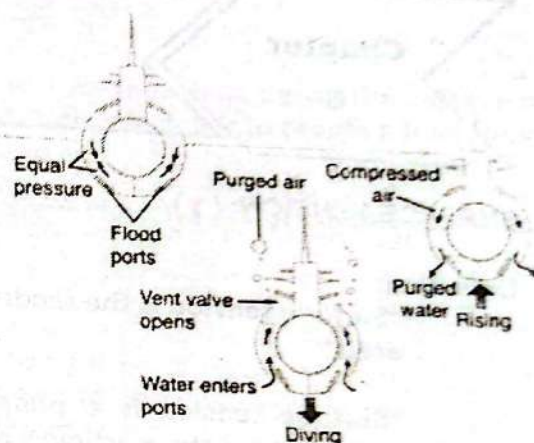
Ships are made of metals such as steel or aluminum alloys. Ships and boats are designed based on the principle of flotation. They carry passengers and goods over water. A ship would sink if its weight, including the weight of its passengers and goods, becomes greater than the upthrust (buoyant force) provided by the water.



• Submarine:

A submarine can travel both on the surface and underwater. It operates based on the principle of flotation. The submarine floats on the water when the weight of the water displaced, which is equal to the volume of the submarine, is greater than its own weight. In this state, it behaves like a ship and remains partially above the water level.

The submarine is equipped with tanks that can be filled with or emptied of seawater. When these tanks are filled with seawater, the weight of the submarine increases. Once its weight exceeds the upthrust (buoyant force), the submarine dives and stays underwater. To resurface, the tanks are emptied of seawater, reducing its weight and allowing it to rise back to the surface.

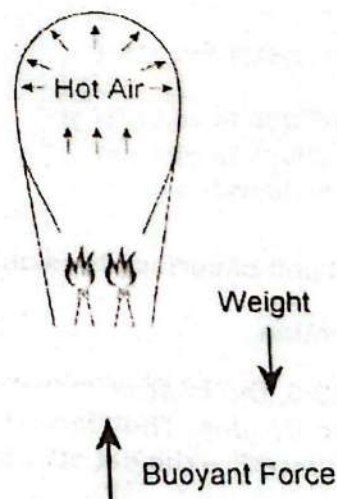


(b) Diving and floating of a submarine

• Hot air balloon:

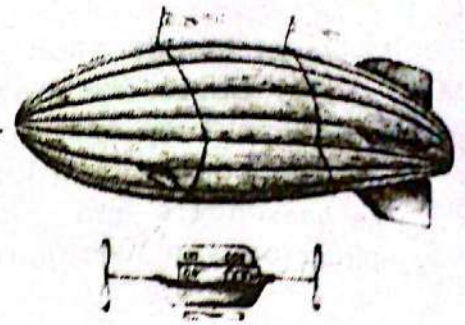
A hot air balloon consists of three parts: the balloon, the burners, and the basket. The burners use propane gas to heat the air inside the balloon. As the air in the balloon heats up, it expands, lowering its density compared to the surrounding air, causing the balloon to rise.

To bring the balloon down, the pilot opens a parachute valve at the top of the balloon, allowing cooler air to enter, which decreases the temperature inside the balloon. This increases the air density, causing the balloon to descend. Alternatively, the pilot can reduce the fuel being burned, allowing the air inside the balloon to cool naturally, which also causes the balloon to come down.



- Air Ship :**

Air ship uses lighter gas such as helium, hydrogen or hot air to rise-up and maneuvering is done by means of propeller attached to it. The lighter gas gives it lifting and the engine is used to propel it.



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Surface Tension

SURFACE TENSION: (γ).

Definition:

"Surface tension is the tendency of a liquid surface at rest to contract, minimizing its surface area."

Or

"Surface tension is a phenomenon where the particles on the surface of a liquid are attracted by the particles beneath them. This attraction causes the surface to contract, minimizing its surface area."

Mathematically it can be defined as

"Force per unit length over which it acts on the surface of liquid perpendicularly is known as surface tension."

$$\text{Surface Tension} = \frac{\text{Force}}{\text{Length}}$$

Or

$$\gamma = \frac{\text{Force}}{\text{Length}}$$

Unit:

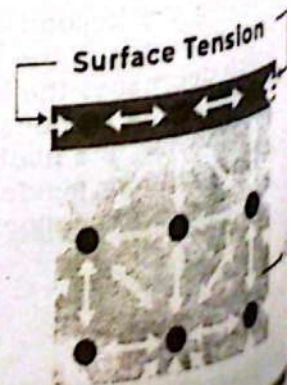
The S.I unit of surface tension is N/m. The c.g.s unit is dynes/cm.

Explanation:

In a liquid, the interior molecules are surrounded by other molecules on all four sides. They face or exhibit an equal attraction from all the sides, therefore the net attracting force acting on it becomes zero.

However the molecules on the Surface are surrounded by the other molecules on three sides as shown in the figure

Therefore this molecule experiences more attraction from the molecules below it. As a result of this, it gets pulled inward and tension is created which is known as surface tension.



Examples of Surface Tension:**1- Drops of water.**

The shape of the drops of water is due to the surface tension. Liquid drops are spherical because a sphere has the smallest surface area for a given volume. The force of gravity pulls down on the drop and therefore the shape of water drop is nearly sphere

2- Needle floating on water:

The steel needle floats on top of the water because of surface tension, which does not allow the needle to sink. If needle is pressed with a finger, the layer of surface tension breaks and the needle sinks in the liquid

3 - Insect walk on water:

Several insects such as the water strider are able to walk on water. Their legs cause the surface of the liquid to become depressed, Which minimizes the potential energy in order to create a balance of forces so the strider can move across the water without breaking it.

Actually insect sinks slightly into the fluid so its weight becomes effective weight of the insect, its true weight less than the buoyant force.

4 - Capillary action: (Capillarity)

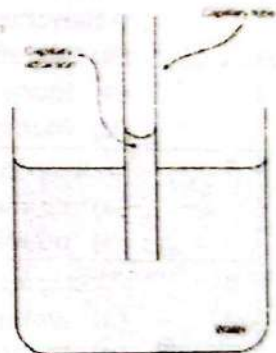
Capillary action is the phenomenon in which a liquid is able to flow through narrow spaces, such as tubes or capillaries, in opposition to external forces like gravity. This remarkable ability arises from a combination of forces, including adhesion and cohesion, as well as surface tension.

Adhesion of water to the walls of a vessel will cause an upward force on the liquid at the edges and result in a meniscus which turns upward. The surface tension act to hold the surface intact.

Capillary action occurs when the adhesion to the walls of the vessel is stronger than the cohesive forces between the liquid molecules.

The height of water column by capillary action is written as

$$h = \frac{2\gamma}{\rho R g}$$



Where,

γ - Surface tension. of the liquid

R - Radius of capillary tube.

ρ - Density of liquid.

g - Gravitational acceleration

5 - Mercury globules on the floor:

Mercury globules are spherical in shape when mercury is dropped over glass surface or floor due to surface tension. The adhesive force between mercury and the other molecules is less than the cohesive force, causing the mercury molecule to maintain its shape.

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Multiple Choice Questions

MCQ'S		ANS.
1.	A completely submerged object always displaces its own (a) weight of fluid (b) volume of a fluid c) density of a fluid d) Area of fluid	b
2.	The pressure exerted on the ground by a man is greatest when: (a) he stands with both feet flat on the ground (b) he stand flat on one foot (c) he stands on the toes of one foot (d) he lies down on the ground	c
3.	In a stationary homogeneous liquid: (a) pressure is the same at all points (b) pressure depends on the direction (c) pressure is independent of any atmospheric pressure on the upper surface of the liquid (d) pressure is the same at all points at the same level	a
4.	One piston in a hydraulic lift has an area that is twice the area of the other. When the pressure at the smaller piston is increased by Δp the pressure at the larger piston: (a) increases by $2\Delta p$ (b) increases by $\Delta p/2$ c) increases by Δp d) increases by $4\Delta p$	c
5.	In a vacuum, an object has (a) no buoyant force (b) no mass c) no weight d) none of these	a
6.	The pressure at the bottom of a pond does NOT depend on (a) water density (b) the depth of the pond c) the surface area of the pond d) None of these	c
7.	A rock suspended by a weighing scale weighs 5N out of water and 3 N when submerged in water. What is the buoyant force on the rock? (a) 3N (b) 5N c) 8N d) 2 N	d
8.	"An object completely submerged in a fluid displaces its own volume of fluid". This is: (a) Pascal's paradox (b) Archimedes' principle c) Pascal's principle d) true, but none of the above	b
9.	Salt water has greater density than freshwater. A boat floats in both freshwater and saltwater. The buoyant force on the boat in salt water is that in freshwater. (a) equal to (b) smaller than c) larger than d) same as	c
10.	You fill a tall glass with ice and then add water to the level of the glass's rim, so some fraction of the ice floats above rim. When the ice melts, what's happens to the water level? (a) the water overflows the rim (b) the water level drops below the rim (c) the water level stays at the rim (d) it depends on the difference in density between water and ice	c

***** Exams Practice MCQs *****

1. Which of the following is NOT a fluid? (a) Water (b) Air (c) Sand (d) Oil	c
2. Which of the following is a property of fluids? (a) They can flow. (b) They can take the shape of their container. (c) They are incompressible. (d) All of the above.	d
3. What is the difference between a liquid and a gas? (a) Liquids have a definite shape, while gases do not. (b) Liquids are less compressible than gases. (c) Liquids have a higher viscosity than gases. (d) All of the above.	b
4. What is the SI unit of pressure? (a) Pascal (Pa) (b) Newton (N) (c) Kilogram (kg) (d) Meter (m)	a
5. What is the difference between a compressible and an incompressible fluid? (a) A compressible fluid can change its volume, an incompressible fluid cannot. (b) A compressible fluid has a higher density than an incompressible fluid. (c) A compressible fluid has a higher viscosity than an incompressible fluid. (d) All of the above.	a
6. Pascal's law states that: (a) Pressure is inversely proportional to the area. (b) Pressure is directly proportional to the volume. (c) Pressure is transmitted equally in all directions through a fluid at rest. (d) Pressure is directly proportional to the temperature.	c
7. Which of the following is NOT an example of Pascal's law in action? (a) A hydraulic press. (b) A car jack. (c) A syringe. (d) A boat floating on water.	d
8. Which of the following is an application of Pascal's law? (a) A hydraulic brake system. (b) A submarine. (c) A water balloon. (d) All of the above.	a
9. Which of the following is TRUE about hydraulic brake systems and Pascal's law? (a) Hydraulic brake systems use Pascal's law to amplify the mechanical advantage of the brake pedal. (b) Hydraulic brake systems use Pascal's law to reduce the mechanical advantage of the brake pedal. (c) Pascal's law has no effect on the mechanical advantage of hydraulic brake systems. (d) Hydraulic brake systems do not use Pascal's law.	a
10. Archimedes' principle states that: (a) The buoyant force acting on an object submerged in a fluid is equal to the weight of the fluid displaced by the object. (b) The pressure at a point in a fluid at rest is equal to the weight of the fluid column above that point. (c) The flow of a fluid through a pipe is proportional to the pressure difference	a

between the ends of the pipe. (d) The drag force on an object moving through a fluid is proportional to the square of the velocity of the object.	
11. Which of the following factors will affect the buoyant force acting on an object? (a) The volume of the object. (c) The gravitational acceleration. (b) The density of the fluid. (d) All of the above.	d
12. A pipe has a diameter of 10 cm and a flow rate of 10 liters per second. What is the velocity of the fluid in the pipe? (a) 2 m/s (b) 20 m/s (c) 10 m/s (d) 1.27 m/s	a
13. According to the law of floatation, an object will float if: (a) The weight of the object is greater than the buoyant force acting on it. (b) The weight of the object is equal to the buoyant force acting on it. (c) The weight of the object is less than the buoyant force acting on it. (d) The weight of object becomes infinite	b
14. Which of the following factors will affect the buoyant force acting on an object? (a) The volume of the object (c) The gravitational acceleration (b) The density of the fluid (d) All of the above	d
15. Surface tension is defined as: (a) The force per unit length acting on a fluid surface. (b) The pressure difference between the inside and outside of a liquid drop. (c) The force required to break a liquid surface. (d) All of the above.	a
16. The SI unit of surface tension is: (a) Newton per meter (Nm^{-1}) (c) Joule (J) (b) Pascal (Pa) (d) None of the above	a
17. Which of the following liquids will have the highest capillary rise in a glass tube? (a) Water (b) Mercury (c) Ethanol (d) Soap solution	d
18. Which of the following factors will decrease the capillary rise of a liquid in a glass tube? (a) Increasing the diameter of the tube (b) Increasing the surface tension of the liquid (c) Decreasing the angle of contact between the liquid and the tube (d) Decreasing the density of the liquid	a
19. A hydraulic system has a piston with an area of 10 cm^2 and a piston with an area of 100 cm^2 . If a force of 100 N is applied to the smaller piston, what force will be produced by the larger piston? (a) 1000 N (b) 100 N (c) 10 N (d) 1 N	a
20. Which of the following is the best explanation for why the drop of olive oil is perfectly spherical? (a) The olive oil is denser than the mixture of alcohol and water. (b) The olive oil is less dense than the mixture of alcohol and water. (c) The surface tension of the olive oil is greater than the surface tension of the	c

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Chapter Formulae & Numericals

Pressure	$P = \frac{F}{A}$
Pascal's Law	$\Delta P = \Delta P_{\text{top}} = \Delta P_{\text{bottom}} = \Delta P_{\text{everywhere.}}$
Hydraulic Lift (Mechanical Advantage)	$\frac{F_2}{F_1} = \frac{A_2}{A_1}$
Archimedes' Principle	APPARENT WEIGHT = ACTUAL WEIGHT - F_{upthrust}
Upthrust Force	$F_{\text{upthrust}} = \rho g V$
Buoyant Force	$F_B \equiv W(\text{displaced fluid})$ $F_B = \rho V g$
Surface Tension	Surface Tension = $\gamma = \frac{\text{Force}}{\text{Length}}$ or $\gamma = \frac{F}{L}$
Capillary action	$h = \frac{2\gamma}{\rho R g}$